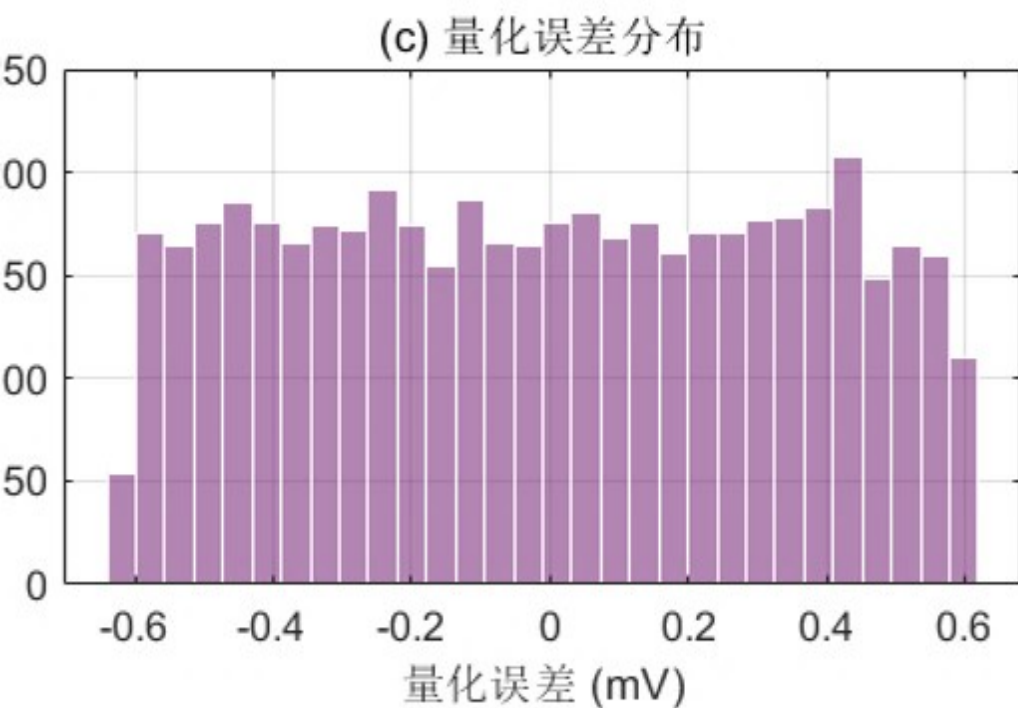
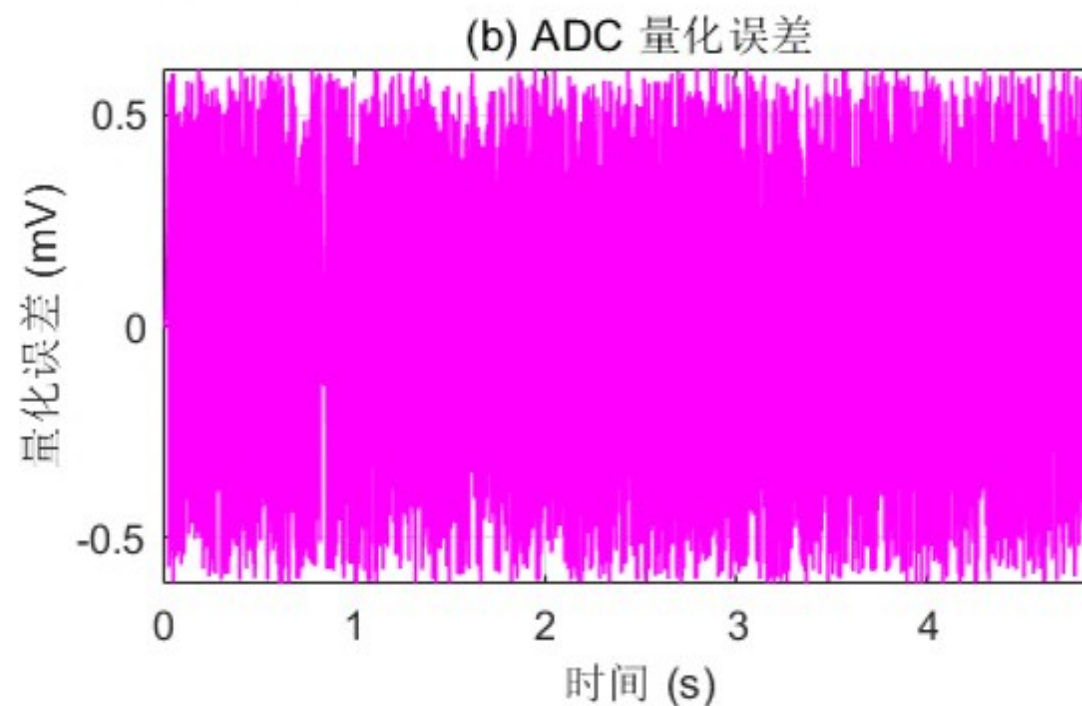
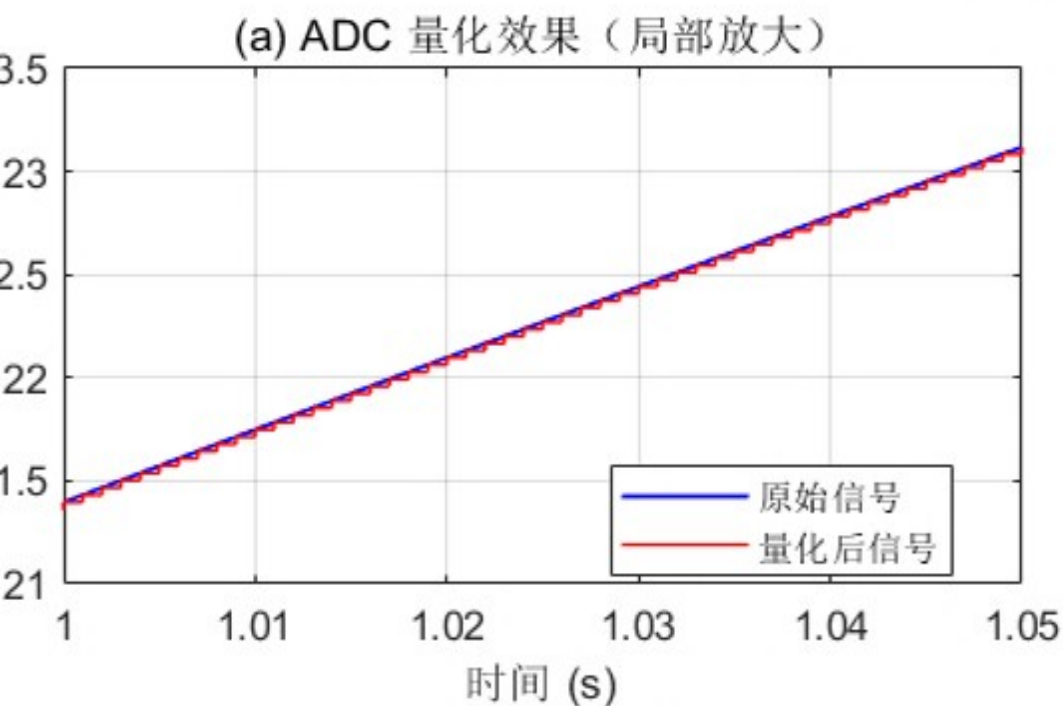


图6: ADC 量化效应分析



ADC 量化参数

ADC 位数: 12 bits

参考电压: 5.00 V

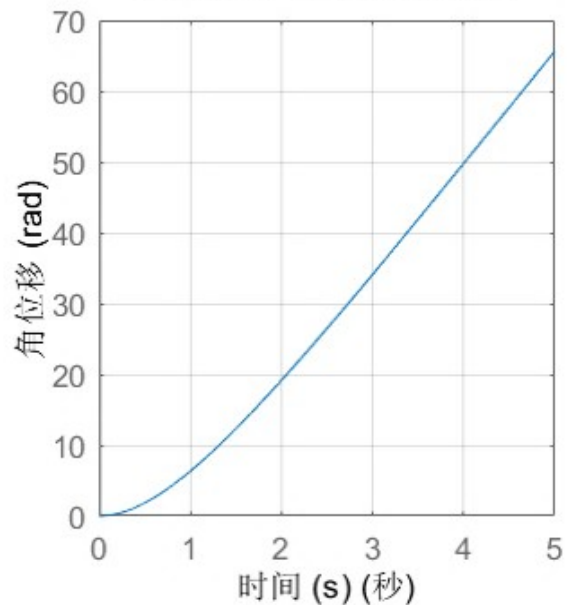
电压分辨率: 1.221 mV

角度分辨率: 0.0070 °

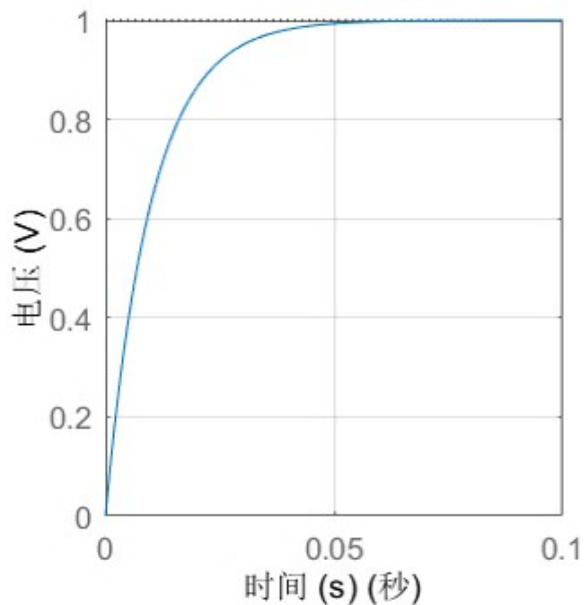
量化噪声 RMS: 0.350 mV

图1: 测试系统各环节阶跃响应

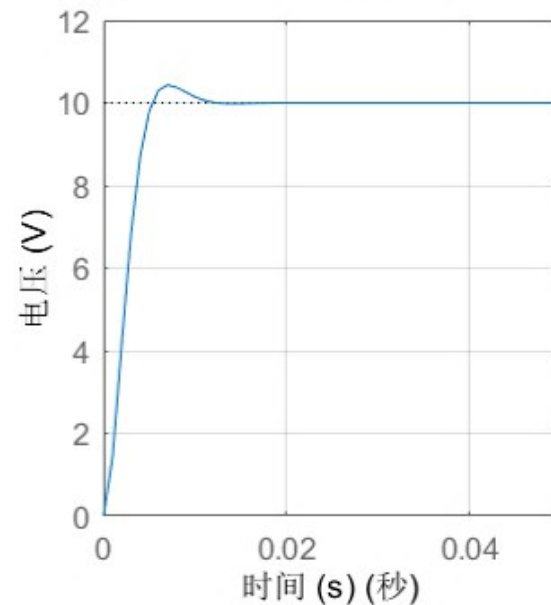
(a) 机械臂 $G1(s)$ 阶跃响应



(b) 传感器 $G2(s)$ 阶跃响应



(c) 信号调理 $G3(s)$ 阶跃响应



(d) 系统总传递函数 $G_{total}(s)$ 阶跃响应

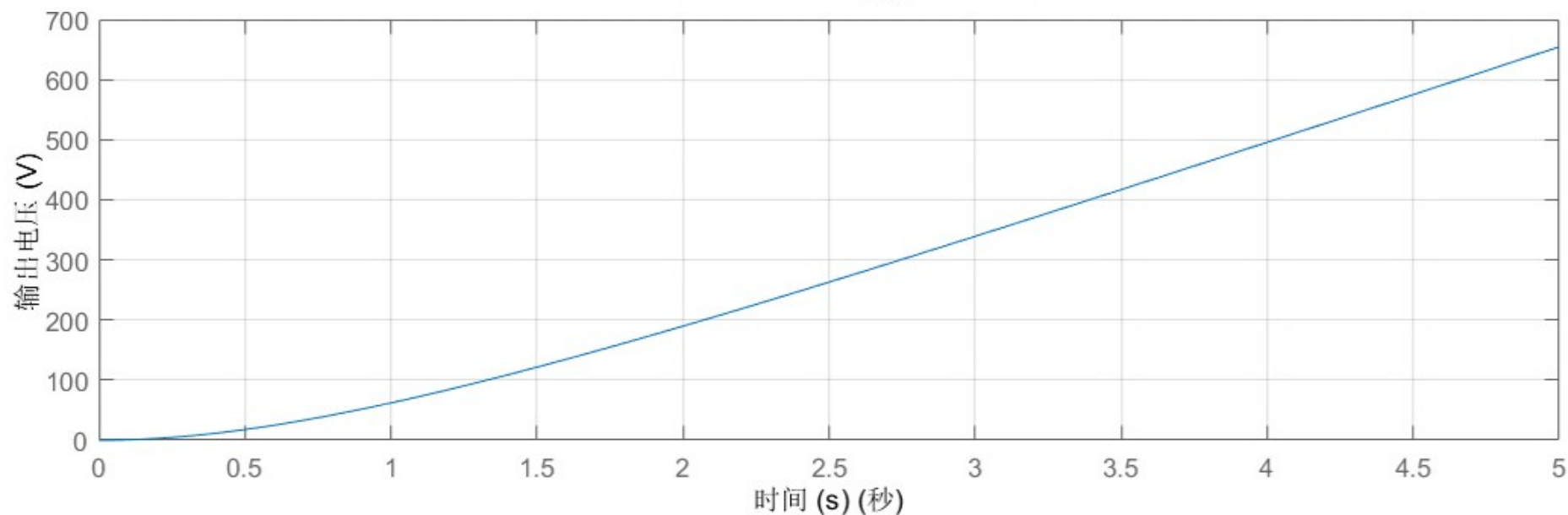


图2. 测试系统频域分析

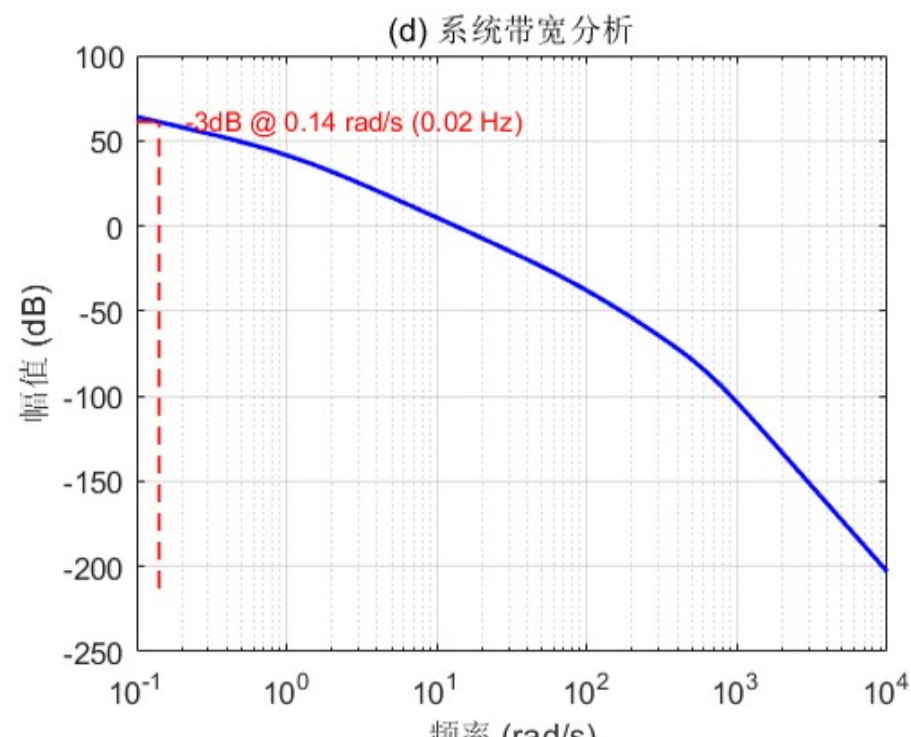
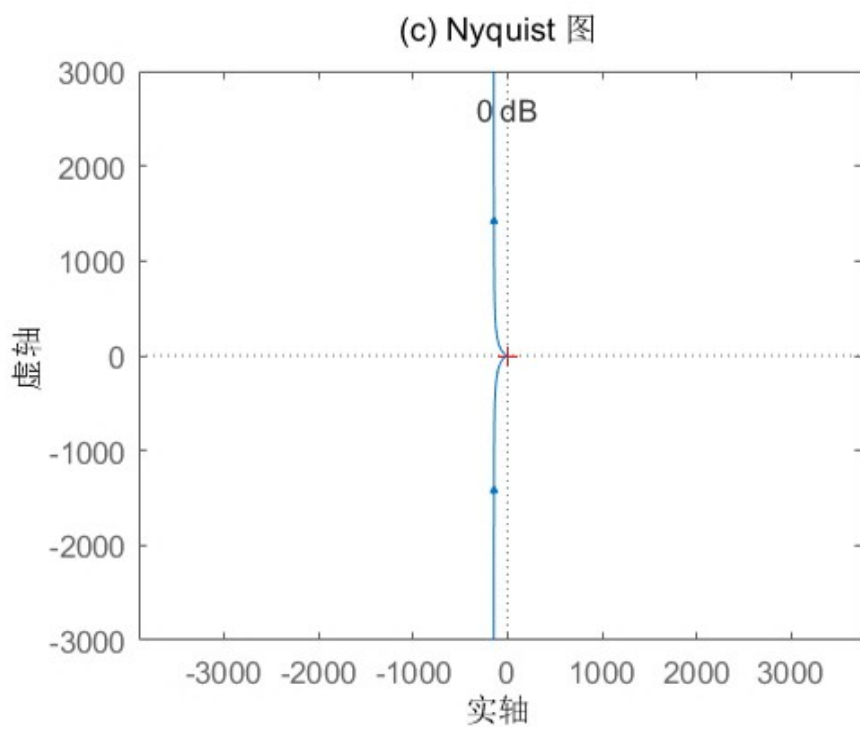
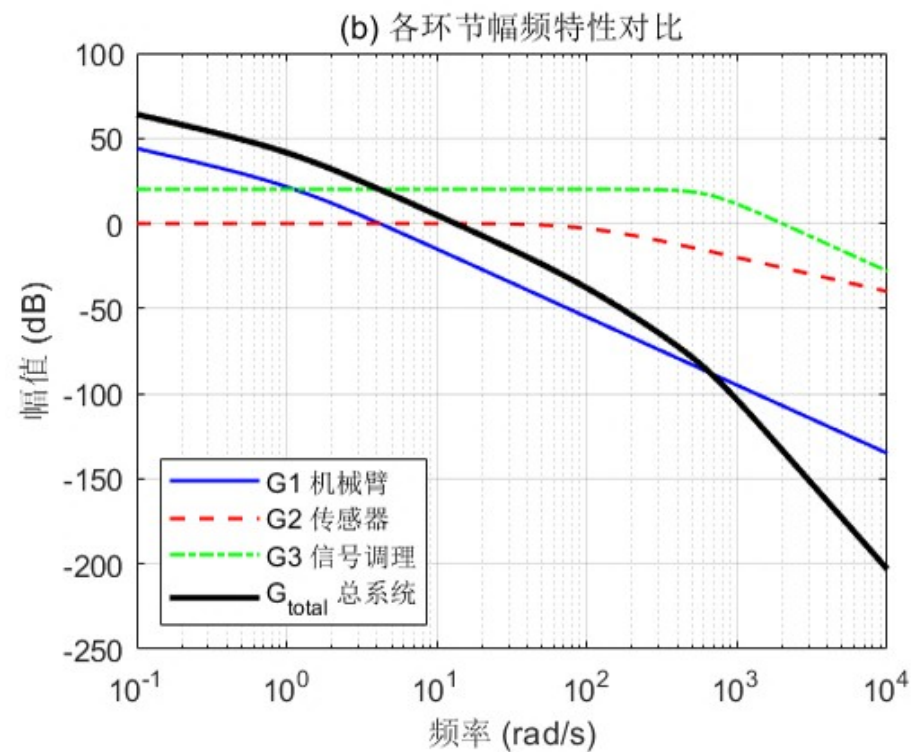
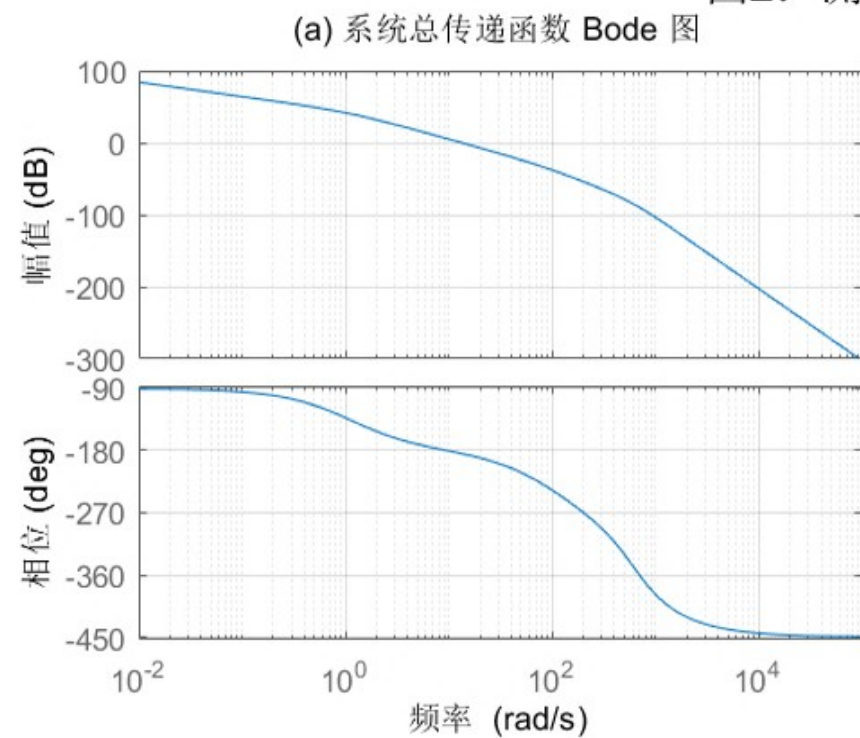


图3: 系统对不同输入信号的时域响应

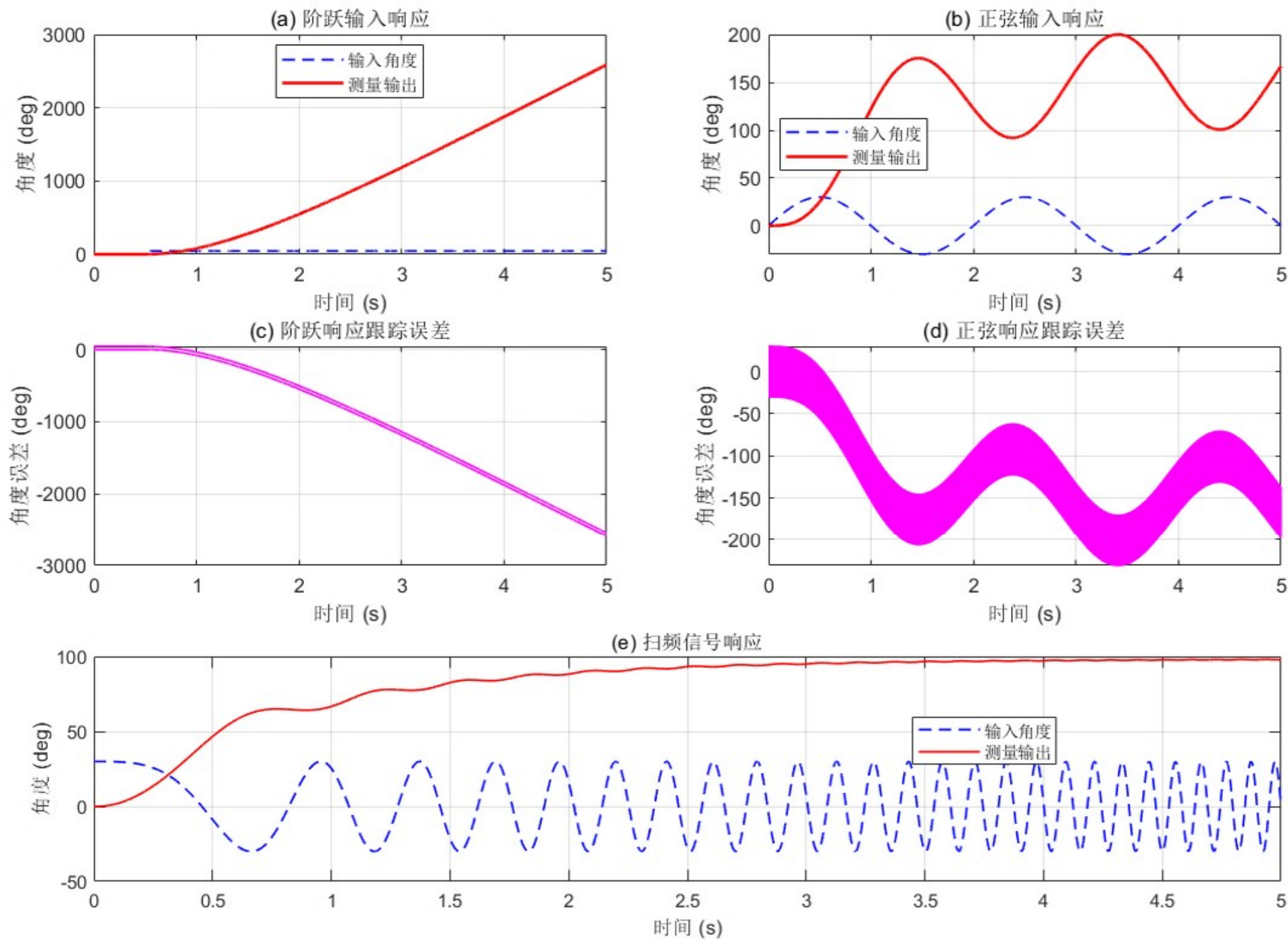


图4：测量噪声对系统的影响分析

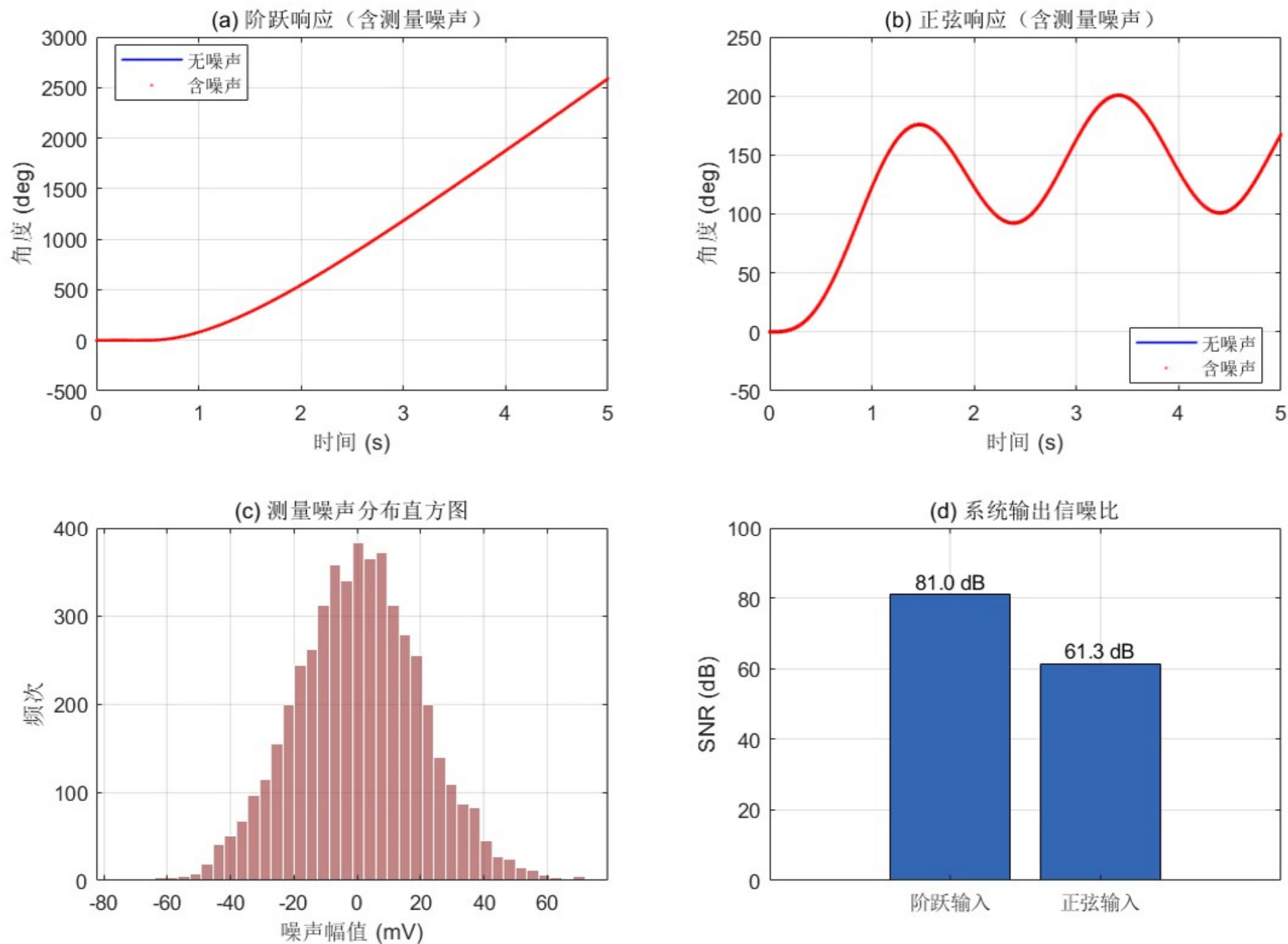
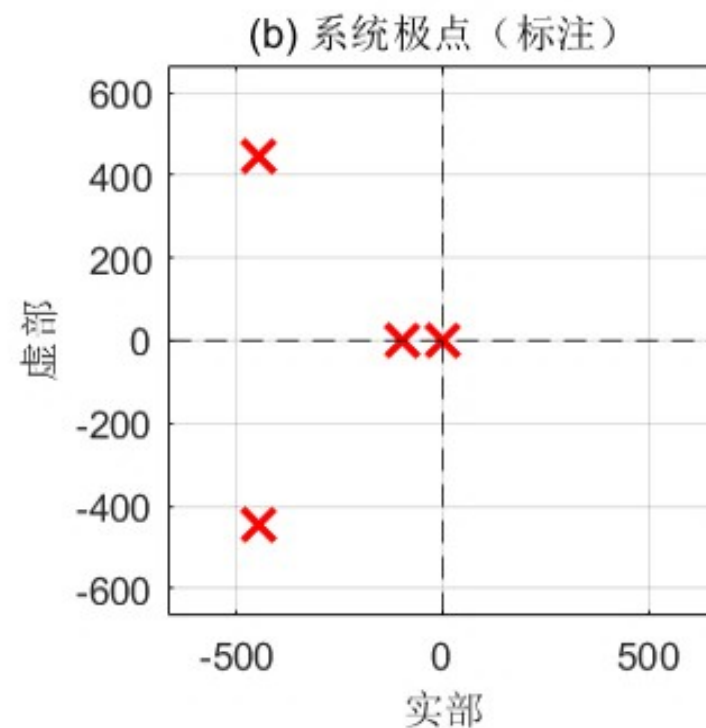
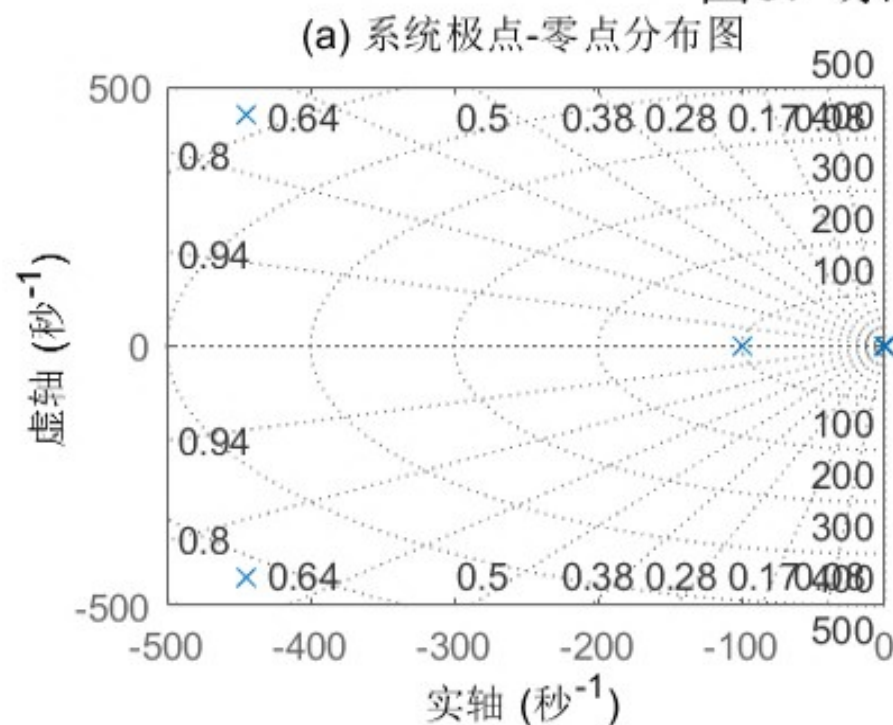


图5: 系统稳定性分析



系统稳定性分析结果

系统极点个数: 5

- 极点 1: $s = 0.0000 + 0.0000j$
- 极点 2: $s = -444.2212 + 444.3554j$
- 极点 3: $s = -444.2212 - 444.3554j$
- 极点 4: $s = -100.0000 + 0.0000j$
- 极点 5: $s = -1.1111 + 0.0000j$

✗ 结论: 系统不稳定! 存在右半平面极点。